Intae Moon

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Personal website: itmoon7.github.io • Google Scholar

RESEARCH PROFILE

I develop machine learning frameworks for high-stakes biomedical and scientific decision-making. My work spans medical foundation models and precision oncology, uncertainty quantification and error-controlled decision-making, applied causal inference, and time-to-event modeling. I am interested in extending these principles to agentic AI systems for reliable multi-step reasoning, scientific discovery, and real-world decision support.

SELECTED RESEARCH CONTRIBUTIONS

ERROR-CONTROLLED DECISION-MAKING FOR FOUNDATION MODELS

Developed an act-or-defer framework that translates foundation-model predictions into actionable decisions while controlling the false discovery rate among acted-upon cases and providing calibrated uncertainty for deferred cases. **I. Moon***, Y. Jin*, and M. Zitnik, Act or Defer: Error-Controlled Decision Policies for Medical Foundation Models, 2026. Under review. *co-first authors.

GENERALIZABLE AI FOR IMMUNOTHERAPY RESPONSE

Contributed to COMPASS, a pan-cancer foundation model that predicts immunotherapy response across cancer types and treatments using biologically grounded immune concepts. W. Shen, **I. Moon**, et al., Nature Medicine, 2026.

AI FOR PRECISION ONCOLOGY

Developed a machine learning framework using routinely collected tumor genomic data to infer cancer origin and evaluate treatment response using a causal-inference framework for cancers of unknown primary. **I. Moon** et al., Nature Medicine, 2023.

LEARNING FROM IRREGULAR LONGITUDINAL CLINICAL DATA

Developed SurvLatent ODE, a Neural-ODE time-to-event framework for irregular longitudinal data with competing risks, applied to cancer-associated venous thromboembolism prediction. **I. Moon**, S. Groha, and A. Gusev, Machine Learning for Healthcare, 2022.

EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY | PhD in Electrical Engineering and Computer Science

Sep 2019 - May 2024 | Cambridge, MA

PhD Thesis: Reliable and trustworthy AI for evidence-based clinical decision support in cancer care

MASSACHUSETTS INSTITUTE OF TECHNOLOGY | SM in Electrical Engineering and Computer Science

Sep 2016 - Sep 2018 | Cambridge, MA • Electrical Energy Conversion and Control

SM Thesis: Design and implementation of a converter with wide operating range using a variable-inverter-rectifier-transformer structure

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN | BS in Electrical and Computer Engineering

May 2016 | Urbana-Champaign, IL • Highest Honors at graduation

RESEARCH EXPERIENCE

ZITNIK LAB | Postdoctoral Research Fellow for Prof. Marinka Zitnik

June 2024 - Present | Boston, MA

- Develop reliable and generalizable machine learning frameworks for biomedical decision-making, spanning error-controlled act-or-defer policies for medical foundation models and foundation-model approaches for precision oncology.

HEALTHY ML LAB | PhD Research Assistant for Prof. Marzyeh Ghassemi

Aug 2022 - May 2024 | Cambridge, MA

- Conducted research on reliable AI models robust to deployment challenges, with a focus on fairness, uncertainty quantification, and clinically meaningful evaluation.

GOOGLE RESEARCH | Student Researcher, Genomics Team in Health AI

June 2022 - Dec 2022 | Cambridge, MA

- Developed a multimodal machine learning framework to learn representations capturing prognostic and biological information from cardiac MRI and genetic data.

GUSEV LAB | PhD Research Assistant for Prof. Alexander Gusev (Harvard Medical School)

May 2019 - May 2024 | Boston, MA

- Developed OncoNPC, a machine learning framework using routinely collected panel sequencing data to predict primary cancer types and support treatment decisions for cancer of unknown primary, as well as Neural-ODE time-to-event models for irregular longitudinal clinical data with competing risks.

POWER ELECTRONICS RESEARCH GROUP | Research Assistant for Prof. David Perreault
Sep 2016 - Jan 2019 | Cambridge, MA

- Designed and built high-frequency power conversion systems with wide operating ranges, including the Variable Inverter-Rectifier-Transformer architecture (US Patent US11716030B2).

PILAWA RESEARCH GROUP | Undergraduate Research Assistant for Prof. Robert Pilawa-Podgurski
May 2015 - Aug 2016 | Champaign, IL

- Designed and built a 1.3 kW power conversion system for data-center power delivery and led the undergraduate team that received the Best Innovation Award at the IEEE International Future Energy Challenge 2016.

PUBLICATIONS

• W. Shen, **I. Moon**, T. H. Nguyen, M. M. Li, Y. Huang, N. Nair, D. Marbach, and M. Zitnik, "Generalizable AI predicts immunotherapy outcomes across cancers and treatments." Nature Medicine, 2026.
• **I. Moon***, Y. Jin*, and M. Zitnik, "Act or Defer: Error-Controlled Decision Policies for Medical Foundation Models." medRxiv, 2026. Under review. *co-first authors.
• **I. Moon**, Y. Jin, and M. Zitnik, "Act or Defer: Error-Controlled Decision Policies for Medical Foundation Models." Proceedings of the Workshop on Structured Data for Health at the 43rd International Conference on Machine Learning, Seoul, South Korea, 2026.
• Y. Huang, X. Su, V. Ullanat, **I. Moon**, I. Liang, L. Clegg, D. Olabode, R. Johnson, N. Ho, M. Gibbs, A. Gusev, B. John, and M. Zitnik, "Multimodal AI predicts clinical outcomes of drug combinations from preclinical data." arXiv, 2025.
• S. Carver, T. Perea-Chamblee, K. Taraszka, **I. Moon**, X. Yu, Y. Ding, J. Carrot-Zhang, and A. Gusev, "Meta-Analysis of Rare Cancers Leveraging Clinically Ascertained Cohorts Reveals Novel Germline Susceptibility Loci." medRxiv, 2026.
• **I. Moon**, H. Jeong, M. Ghassemi, and A. Gusev, "Robust and fair time-to-event framework for predicting cancer-associated venous thromboembolism (VTE) using routinely-collected clinical and panel-sequencing data." American Society of Human Genetics (ASHG), Washington, D.C., 2023.
• G. Lee, A. Li, **I. Moon**, D. Katritsis, Y. Pantos, F. Stingo, D. Fabbrico, R. Molinaro, F. Taraballi, W. Tao, and C. Corbo, "Protein Corona Sensor Array Nanosystem for Detection of Coronary Artery Disease." Small, 2023.
• **I. Moon** and A. Gusev, "Machine learning for improved clinical management of cancers of unknown primary." Nature Medicine, Research Briefing, 2023.
• **I. Moon**, J. LoPiccolo, S. Baca, K. Kehl, M. Hassett, D. Liu, D. Schrag, and A. Gusev, "Machine learning for genetics-based classification and treatment response prediction in cancer of unknown primary." Nature Medicine, 2023. **[Featured on MIT News, DFCI News, and Nature Portfolio genetics research.]**
• A. Ayed, C. Saillard, J. Onofrey, **I. Moon**, S. Chang, A. Feldman, and M. Nayan, "Machine learning to predict future PSA in patients with prostate cancer managed with active surveillance." Journal of Clinical Oncology, 2023.
• **I. Moon**, J. LoPiccolo, S. Baca, K. Kehl, and A. Gusev, "Utilizing Electronic Health Records (EHR) and tumor panel sequencing to demystify prognosis of Cancer of Unknown Primary (CUP) patients." **[Top 10% Reviewers' Choice Abstract, American Society of Human Genetics (ASHG), Los Angeles, CA, 2022.]**
• H. Jeong, K. Yang, Z. Wei, Y. Ma, **I. Moon**, and S. Kanjilal, "Estimating the Treatment Effect of Antibiotics Exposure on the Risk of Developing Anti-Microbial Resistance." Women in Machine Learning Workshop (WiML), NeurIPS 2022.
• **I. Moon**, S. Groha, and A. Gusev, "SurvLatent ODE: A Neural ODE based time-to-event model with competing risks for longitudinal data improves cancer-associated venous thromboembolism (VTE) prediction." Proceedings of Machine Learning Research (PMLR), Machine Learning for Healthcare, 2022.
• **I. Moon**, S. Chakraborty, and D. J. Perreault, "A High-performance 65 W Universal ac-dc Converter Using a Variable-Inverter-Rectifier-Transformer with Improved Step-down Capability." IEEE Energy Conversion Congress and Exposition (ECCE), 2019.
• **I. Moon**, M. K. Ranjram, S. Chakraborty, and D. J. Perreault, "A Wide Operating Range Converter Using a Variable-Inverter-Rectifier-Transformer with Improved Step-Down Capability." IEEE Applied Power Electronics Conference and Exposition (APEC), 2019.
• M. K. Ranjram, **I. Moon**, and D. J. Perreault, "Variable-Inverter-Rectifier-Transformer: A Hybrid Electronic and Magnetic Structure Enabling Adjustable High Step-Down Conversion Ratios." IEEE Transactions on Power Electronics, 2018. **[Highlighted paper of the issue].**
• Y. Lei, C. Barth, S. Qin, W. Liu, **I. Moon**, A. Stillwell, D. Chou, T. Foulkes, Z. Ye, Z. Liao, and R. C. N. Pilawa-Podgurski, "A 2-kW Single-Phase Seven-Level Flying Capacitor Multilevel Inverter With an Active Energy Buffer." IEEE Transactions on Power Electronics, 2017.
• **I. Moon**, C. F. Haken, E. K. Saathoff, E. Bian, Y. Lei, S. Qin, D. Chou, S. Sedig, W. Chung, and R. C. N. Pilawa-Podgurski, "Design and implementation of a 1.3 kW, 7-level flying capacitor multilevel AC-DC converter with power factor correction." IEEE Applied Power Electronics Conference and Exposition (APEC), 2017.
• S. Qin, Y. Lei, **I. Moon**, C. Haken, E. Bian, E. Saathoff, W. Chung, D. Chou, and R. C. N. Pilawa-Podgurski, "A high power density power factor correction front end based on a 7-level flying capacitor multilevel converter." IEEE Southern Power Electronics Conference (SPEC), 2016.
• C. B. Barth, **I. Moon**, Y. Lei, S. Qin, and R. C. N. Pilawa-Podgurski, "Experimental evaluation of capacitors for power buffering in

single-phase power converters." IEEE Energy Conversion Congress and Exposition (ECCE), 2015.

INVITED TALKS

• "Trustworthy AI for Precision Oncology," Invited Seminar, School of Medicine and Department of Artificial Intelligence, Sungkyunkwan University (SKKU), Suwon, South Korea, July 24, 2026.

• "Trustworthy AI for Precision Oncology," Medical Intelligence Lecture Series, Institute for AI and Social Innovation, Yonsei University, Seoul, South Korea, July 29, 2026.

• "Machine learning for genetics-based classification and treatment response prediction in cancer of unknown primary," Molecular Tumor Board, Dana-Farber Cancer Institute, Boston, MA, 2023.

• "Time-to-event model with competing events for irregularly sampled time-series," Cipherome, Inc., San Jose, CA, 2022.

• "A Wide Operating Range Power Converter Using a Variable-Inverter-Rectifier-Transformer with Improved Step-Down Capability," Center for Integrated Circuits and Systems (CICS) Review, Massachusetts Institute of Technology, 2018.

• "Miniaturized Grid-tied ac-dc Power Converters with Wide Output Voltage Ranges," Texas Instruments, Manchester, NH, 2018.

• "A 1.3 kW, 7-level Flying Capacitor Multilevel AC-DC converter for a Data Center Power Delivery," IEEE International Future Energy Challenge, National Taiwan University, 2016.

PROFESSIONAL SERVICE

AREA CHAIR Conference on Health, Inference, and Learning (CHIL)

- Models and Methods Area Chair, CHIL 2025.
- Applications and Practice: Investigation, Evaluation, Interpretation, and Deployment Area Chair, CHIL 2026.

JOURNAL REVIEWER Nature Portfolio

• Communications Biology (2024) • Communications Medicine (2024) • npj Precision Oncology (2026)

CONFERENCE AND WORKSHOP REVIEWER • ML4H Symposium (2023) • ICLR TS4H Workshop (2024) • CHIL (2024) • MLHC (2024, 2026) • RECOMB (2025, 2026) • NeurIPS TS4H Workshop (2025) • ICML SD4H Workshop (2026)

RESEARCH COMMUNITY LEADERSHIP 2021 | Longwood Medical Area, Boston, MA

- Organized a causal inference reading group across research laboratories in the Longwood Medical Area; materials available here.

TEACHING AND MENTORING

DR. SUSANNE E. CHURCHILL SUMMER INSTITUTE IN BIOMEDICAL INFORMATICS (SIBMI) | Research Mentor, Harvard Medical School

Boston, MA

- Mentored an undergraduate student on using large language models with molecular data to infer actionable molecular alterations under limited panel sequencing.

6.871 MACHINE LEARNING FOR HEALTHCARE | Graduate Teaching Assistant working with Prof. David Sontag

Jan 2022 - June 2022 | Cambridge, MA

- Developed weekly recitation content and Python-based problem sets, coordinated clinical research collaborations for student projects, and mentored project teams. Received the Carlton E. Tucker Award for teaching excellence (overall instructor rating 6.8/7.0).

6.004 COMPUTATION STRUCTURES | Graduate Teaching Assistant

Sep 2018 - May 2019 | Cambridge, MA

- Helped develop laboratory assignments and course materials in RISC-V assembly and Bluespec and guided students in building pipelined processors and understanding computer architecture concepts.

AWARDS

CHARLES J. EPSTEIN TRAINEE AWARDS FOR EXCELLENCE IN HUMAN GENETICS RESEARCH | Predoctoral Semifinalist

Aug 2022 | American Society of Human Genetics (ASHG), Los Angeles, CA

- Selected as one of 60 semifinalists from more than 700 applicants.

CARLTON E. TUCKER AWARD FOR TEACHING EXCELLENCE | Annual department teaching award, MIT Electrical Engineering and Computer Science

Jun 2022 | Cambridge, MA

BEST INNOVATION AWARD | IEEE International Future Energy Challenge 2016 (Team Leader)
Jul 2016 | National Taiwan University, Taipei, Taiwan

RUSSELL E. BERTHOLD SCHOLARSHIP | Power and Energy Systems Scholarship
Nov 2015 | Champaign, IL

GOOGLE LITTLE BOX CHALLENGE | Finalist
Oct 2015 | Champaign, IL

SKILLS

MACHINE LEARNING & COMPUTING

• PyTorch • TensorFlow • JAX • Hugging Face
• Python • SQL • Slurm/HPC • scikit-learn • R
• C/C++ • MATLAB • LaTeX

LANGUAGES

Native fluency:
English, Korean
Basic fluency:
French